

KIAN ENTEZARI

Computer Vision | Embedded Systems | Robotics | Biosensors

Portfolio | (925) 808-1559 | kianxyz@gmail.com

SUMMARY

Electrical and Computer Engineering student at UC Berkeley with a 4.00 GPA and hands-on experience in computer vision, embedded systems, and biosensor development. Built a stereo vision pipeline for a commercial robotic arm at Mantis Robotics and conducting independent research on a low-cost electrochemical potentiostat for non-invasive sweat analysis.

EDUCATION

University of California, Berkeley

Expected Spring 2028

B.S., Electrical and Computer Engineering | GPA: 4.00 | 95 units completed

TECHNICAL SKILLS

Languages: Python, C, C++, C#

Tools: AutoCAD, Git, ROS 2 | Planned Spring 2027: PyTorch

Perception: DIAVision, OpenCV, YOLOv8, ONNX Runtime, LiDAR

Protocols: Modbus TCP, Google API, Meta API, OpenAI API

Electronics: KiCad, LTspice, potentiostat design

Spoken Languages: Spanish (conversational)

HDLs: Planned Spring 2027: Verilog, SystemVerilog

Embedded Platforms: ESP32, Arduino

EXPERIENCE

Perception Engineer Intern | Mantis Robotics

Nov 2025 – Aug 2026

- Developed a stereo vision pipeline for the MR-1 robotic arm achieving multi-object 3D localization with <1 mm accuracy, while cutting hardware costs by over 90% by replacing industrial cameras with two low-cost webcams.
- Integrated YOLOv8 object detection (OpenCV, ONNX Runtime in C#) for real-time object detection and class identification.
- Implemented camera-to-robot coordinate transformations for robotic object localization.
- Built a LiDAR-camera fusion pipeline using YOLO detections and projected point clouds for 3D object localization.
- Created a Modbus TCP server to transmit robot coordinates from the vision pipeline to the robot controller.
- Integrated and tested digital I/O for robotic hardware communication and control.

Software Engineer Intern | Inner Health & Wellness

Oct 2025 – Jan 2026

- Built and deployed an AI chatbot into the company's app using the OpenAI API, giving clients an in-app resource for health and wellness guidance.
- Developed a backend server and data ingestion workflow using the Google API to centralize company content and deliver it in-app.
- Integrated the Meta API for in-app messaging and deployed all services on Vercel.

Tutor | Mathematics Tutoring

Jan 2026 – Present

- Tutor Calculus through Differential Equations, adapting to individual learning styles and pacing.

PROJECTS

Independent Research | Multi-Analyte Electrochemical Biosensor

May 2026 – Present

- Designing and building a low-cost TLV272-based potentiostat circuit capable of measuring blood sugar, lactate, and other analytes (cortisol, blood alcohol, vitamin C) from sweat via electrochemical detection.
- Developing a reverse iontophoresis system for non-invasive interstitial fluid extraction, enabling continuous monitoring without skin puncture.